

Hybrid Extractive-Abstractive Summarization of Financial Narratives via Unsupervised Semantic Clustering and DistilBERT Scoring

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Abstract—Summarizing annual financial reports, such as Form 10-K, presents a unique challenge due to their extreme length—often exceeding 50,000 tokens—and the dispersed nature of salient information, which confounds standard truncation-based strategies. While recent benchmarks like FINDSum have enabled sophisticated end-to-end models, these methods require massive computational resources unattainable for many researchers.

This paper proposes a resource-constrained “Cluster-Rank-Abstract” pipeline that combines unsupervised semantic clustering with parameter-efficient supervised scoring to overcome the input truncation bottleneck. We granularize documents into sentences embedded by a lightweight MiniLM model and partition them into 30 latent topics using K-Means clustering. To refine content selection, we employ a Reference-Guided Density strategy to identify high-value topics, followed by a Weighted Decay algorithm that dynamically allocates sentence budgets based on cluster importance. Finally, the extracted content is synthesized into a coherent summary by a Longformer Encoder-Decoder (LED) fine-tuned on a randomly sampled 17% subset of the data.

Experimental results on a single NVIDIA T4 GPU demonstrate that our pipeline achieves a ROUGE-1 score of 37.76, outperforming a standard BART baseline by 87.8% and retaining nearly 70% of the performance of state-of-the-art systems while using a fraction of the training data. This establishes the viability of density-based content selection as a “Green AI” alternative for long-document financial summarization.

Index Terms—Financial Summarization, NLP, K-Means, DistilBERT, Longformer

I. INTRODUCTION

A. The Challenge of Financial Information Overload

In the modern financial landscape, market participants face a critical bottleneck: the sheer volume of unstructured data. Timely decision-making is essential, yet the primary sources of information—annual financial reports (Form 10-K)—are notoriously dense [2]. These documents often exceed 180 pages or 50,000 tokens, containing a complex mix of quantitative data, executive narratives, and legal risk disclosures [3]. To navigate this data efficiently, analysts and investors require automated tools capable of distilling these massive documents into concise, actionable summaries [4].

B. Limitations of Current Approaches

Automatic Text Summarization (ATS) systems generally fall into two categories, both of which face significant limitations when applied to long-form financial documents.

a) Extractive Methods.: These systems function by selecting and concatenating the most statistically significant sentences from the source text, often using graph-based ranking algorithms like TextRank [5]. While this preserves the factual accuracy of the original text, the resulting summaries often lack coherence, suffering from disjointed narrative flows and a lack of connective transitions [6].

b) Abstractive Methods.: Neural sequence-to-sequence models (e.g., Transformers) generate new text to capture the core meaning, much like a human writer [7]. While these produce more fluent and readable summaries, standard models have restricted context windows (typically 512 to 1,024 tokens) [8]. When applied to a 50,000-token financial report, these models are forced to truncate the vast majority of the document, leading to severe information loss—a phenomenon known as “forgetting” [3].

C. Proposed Solution: A Resource-Efficient Hybrid Pipeline

To address the “truncation bottleneck” without requiring the massive computational resources typically needed for end-to-end training on long documents, we propose a modular “Cluster-Rank-Abstract” framework.

Our approach combines the structural awareness of unsupervised learning with the precision of supervised scoring. Specifically, we utilize K-Means clustering on SentenceBERT embeddings to map the document’s latent topics, ensuring global coverage [9]. We then refine this selection using a parameter-efficient DistilBERT scorer to identify high-value content, which is finally synthesized by a Longformer Encoder-Decoder (LED) [10]. This pipeline offers a viable, low-resource alternative for summarizing extensive financial narratives.

D. Paper Structure

The remainder of this paper is organized as follows: Section II reviews the related work in automatic summarizations specifically focusing on long document and financial domains. Section III details our proposed methodologies including the mathematical formulation of our weighted decay extraction algorithm. Section IV describes the experimental setups dataset statistics, and evaluation metrics. Section V presents the quantitative results, ablation studies and the qualitative analysis of generated summaries. Finally Section VI concludes the

paper with a summary of findings limitations and directions for future work.

II. RELATED WORK

A. The Unique Challenges of Financial Text

Financial summarization differs significantly from general news summarization due to the extreme length and specialized lexicon of the source documents. Loughran and McDonald [11] established that standard dictionaries often misclassify financial tone, necessitating domain-specific handling to differentiate between standard accounting terminology and genuine risk factors. To address the scarcity of data in this domain, [1] introduced FINDSum, a large-scale dataset of annual reports (Form 10-K) that highlights the difficulty of summarizing documents where salient information is scattered across tens of thousands of tokens.

Furthermore, annual reports (Form 10-K) are notoriously dense; El-Haj *et al.* [12] highlighted in the Financial Narrative Summarisation (FNS) shared task that these documents contain “scattered” salient information—key risks may appear in the introduction but be detailed dozens of pages later. This dispersion of information makes simple truncation strategies ineffective, as they inevitably discard critical narrative elements.

B. Baseline Approaches: Extractive vs. Abstractive

Early approaches to summarization were predominantly extractive, relying on graph-based algorithms such as LexRank [13] to identify central sentences using eigenvector centrality. While computationally efficient and factually safe, these methods often produce disjointed summaries that lack narrative flow.

To improve content coverage, recent work has leveraged unsupervised clustering; Miller [14] demonstrated that K-Means clustering on BERT embeddings can effectively reduce redundancy by selecting representative sentences from diverse topic groups.

Conversely, neural sequence-to-sequence models have revolutionized abstractive summarization. Models like BART [15] generate fluent, human-like text through denoising-based pre-training. However, standard Transformer architectures suffer from quadratic memory complexity with respect to sequence length, typically limiting inputs to 512 or 1,024 tokens. Sanh *et al.* [20] introduced DistilBERT, a compressed and computationally efficient version of BERT, but even lightweight models face the “truncation bottleneck” when processing 50,000-token financial documents, resulting in severe loss of context.

C. Long Document Solutions and Efficient Attention

To address the limitations of standard Transformers, researchers have proposed specialized architectures for long sequences. Zaheer *et al.* [16] introduced Big Bird, which employs sparse attention patterns to reduce complexity from quadratic to linear, thereby enabling substantially longer input windows.

Similarly, Huang *et al.* [17] developed Efficient Attentions, demonstrating that head-wise positional strides can effectively capture salient information in long-form government reports without the memory overhead of dense attention.

Earlier work by Cohan *et al.* [22] proposed a Discourse-Aware Attention Model that utilizes a hierarchical encoder to represent the structure of scientific documents. While these models improve long-context handling, many remain computationally expensive or require extensive fine-tuning on large-scale hardware, limiting their accessibility.

D. Hallucination and the “Green AI” Imperative

A major limitation of abstractive models in high-stakes settings is the risk of hallucination—producing content that is fluent but factually unsupported. Maynez *et al.* [18] showed that modern abstractive systems frequently generate statements not grounded in the source text, a critical flaw for domains such as finance.

Recent work by Zhang *et al.* [19] indicates that mitigating hallucination requires domain-conditional mutual information constraints to ensure faithful generation.

In addition to accuracy concerns, the computational and environmental cost of training large Transformer models has prompted a movement toward “Green AI.” Schwartz *et al.* [21] argue that the research community should prioritize efficiency alongside performance, calling for approaches that avoid exponential increases in computational cost.

Our proposed lightweight hybrid pipeline aligns with this philosophy by balancing the global coverage of clustering with the fluency of abstractive generation in a resource-efficient manner.

III. METHODOLOGY

A. Overview of the “Cluster-Rank-Abstract” Framework

To overcome the computational constraints of processing 50,000-token financial reports on a single GPU, we propose a modular pipeline that filters information based on semantic density. Unlike standard truncation strategies which indiscriminately cut text after a fixed limit, our approach creates a semantic map of the document to select high-value sentences from diverse topics before generation.

The pipeline consists of three stages: (1) Unsupervised Semantic Clustering, (2) Reference-Guided Density Ranking, and (3) Long-Context Abstractive Synthesis via LED.

B. Dataset and Preprocessing

We utilized the **FINDSum** dataset introduced by Liu *et al.* [1], which contains comprehensive annual reports (Form 10-K).

- **Data Merging:** The original dataset separates “Results of Operations” (ROO) and “Liquidity” into distinct summarization tasks. To simulate a realistic, holistic analyst workflow where these narratives are intertwined, we merged these subsets into a composite dataset.
- **Preprocessing:** Raw text was cleaned of HTML artifacts and boilerplate legal disclaimers. We utilized the NLTK

Financial Summarization Pipeline

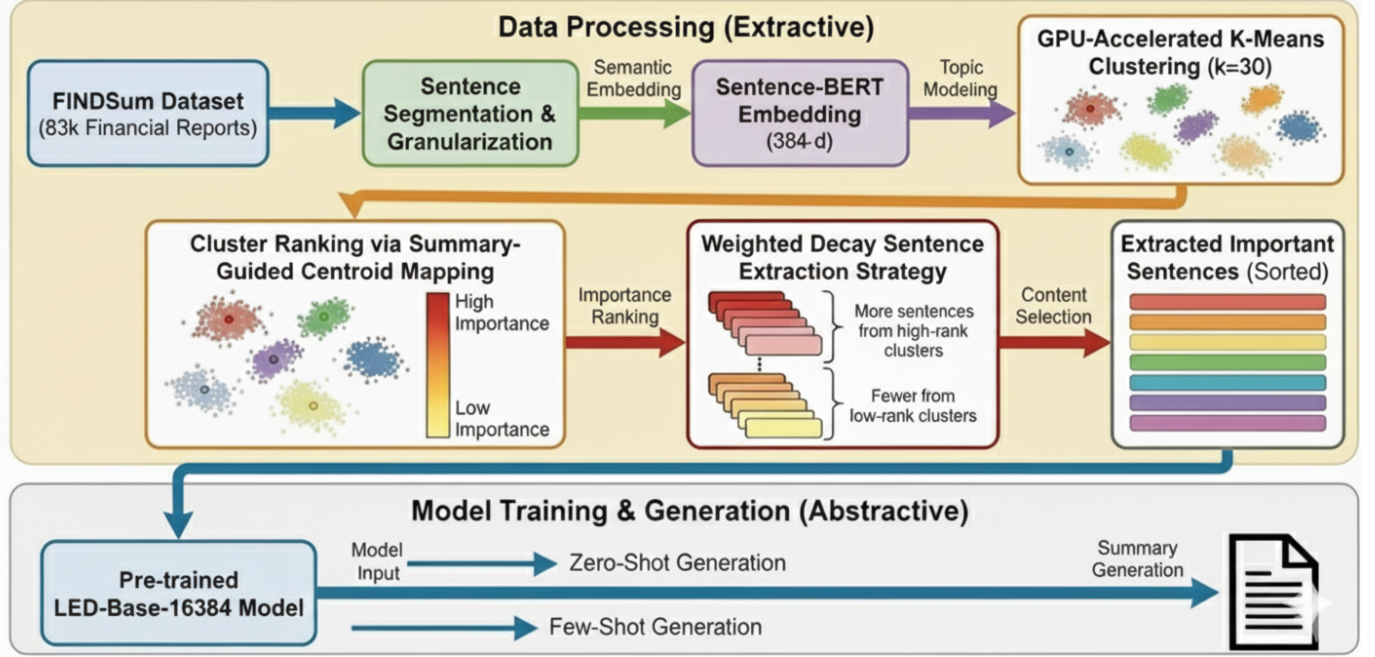


Fig. 1. Workflow of the Proposed Summarization System. The process begins with preprocessing 83k financial reports. It employs a “filter then generate” approach where salient information is first extracted using semantic clustering and importance ranking. The extracted content is then fed into a Longformer Encoder-Decoder (LED) model to generate the final abstractive summary.

sent_tokenize library to segment the continuous document text into atomic sentences

$$S = \{s_1, s_2, \dots, s_n\}.$$

C. Stage 1: Unsupervised Semantic Clustering

To create a semantic map of the document, we require a vector representation for every sentence. To maintain our “resource-efficient” design constraints, we utilized the all-MiniLM-L6-v2 model from the Sentence-Transformers library [9]. This model maps sentences to a 384-dimensional dense vector space optimized for high semantic performance and low memory usage.

We applied **K-Means clustering** on these MiniLM embeddings to partition the document into $K = 30$ latent topics (clusters),

$$C = \{C_1, C_2, \dots, C_{30}\},$$

where each centroid μ_j represents a distinct thematic section of the 10-K report (e.g., “Risk Factors,” “Market Analysis,” “Legal Proceedings”).

D. Stage 2: Reference-Guided Cluster Ranking & Selection

A critical challenge in long-document summarization is identifying which clusters contain the most salient information. We adopt a **Reference-Guided Density** strategy to determine cluster importance.

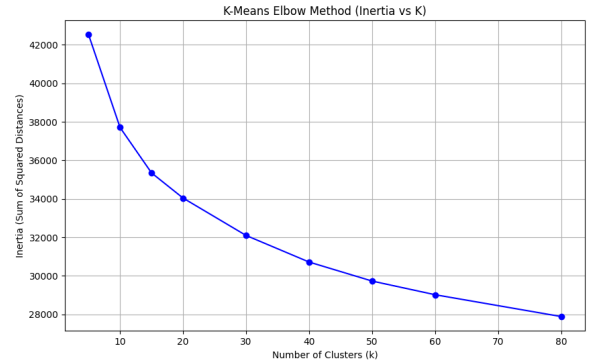


Fig. 2. Determination of Optimal Cluster Count via the Elbow Method. The graph illustrating the change in within-cluster sum of squares (inertia) as the number of clusters k increases from 5 to 80. The “elbow” of the curve suggests an optimal cluster number where the marginal gain in variance explanation decreases, justifying the selection of $k = 30$.

1) *Mapping Reference Summaries:* We processed the ground-truth reference summaries using the same pipeline. Each reference summary sentence was encoded and mapped to the nearest cluster centroid μ_j using Euclidean distance.

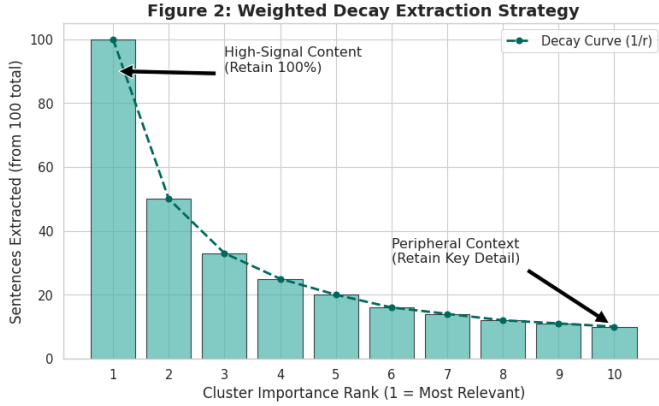
2) *Density-Based Ranking:* The *Importance Score* I_j of a document cluster is defined as the number of reference sentences mapped to it. Clusters were then ranked from $r = 1$ (highest density) to $r = 30$.

Algorithm 1 Weighted Decay Extraction Strategy

Require: Document Clusters $C = \{c_1, \dots, c_k\}$, Importance Ranks $R = \{r_1, \dots, r_k\}$

Ensure: Extracted Sentences E

```
1:  $E \leftarrow \emptyset$ 
2: for each cluster  $c_i$  in order of increasing rank do
3:    $N_{\text{total}} \leftarrow |c_i|$ 
4:    $r_i \leftarrow \text{rank of } c_i$ 
5:    $N_{\text{keep}} \leftarrow \max(1, \lfloor N_{\text{total}} \cdot (1/r_i) \rfloor)$ 
6:   Sort sentences in  $c_i$  by DistilBERT score (descending)
7:   Select top  $N_{\text{keep}}$  sentences as  $S_{\text{selected}}$ 
8:    $E \leftarrow E \cup S_{\text{selected}}$ 
9: end for
10: return  $E$  sorted by original document order
```



3) *Weighted Decay Extraction Strategy*: While clusters are ranked by importance, individual sentence selection uses a fine-tuned **DistilBERT** classifier.

Selecting a uniform number of sentences per cluster would overweight high-density clusters and ignore smaller but relevant ones. To solve this, we use a **Weighted Decay Extraction Strategy**, summarized in Algorithm 1.

E. Stage 3: Long-Context Abstractive Generation

The extracted sentences contain high-value content but lack coherence. To synthesize them into a fluent narrative, we utilize the **Longformer Encoder-Decoder (LED)** [10].

1) *The Longformer Advantage*: Traditional Transformer models such as BART are limited to a 1,024-token context window. LED extends this capability to **16,384 tokens** via sparse local/global attention. This allows the model to process our extracted “semantic digest” instead of the full 50,000-token report.

2) *Efficient Fine-Tuning*: To adapt LED to financial text efficiently, we fine-tuned the model on a **17% randomly sampled subset** of the FINDSum dataset. Because Stage 2 delivers curated, high-precision content, the LED requires far less training data to produce coherent, professional-grade summaries.

LED was fine-tuned using standard seq2seq objectives with a learning rate of 3×10^{-5} and a beam size of 4.

IV. EXPERIMENTAL SETUP

A. Dataset and Sampling Details

We evaluated our pipeline on the **FINDSum** dataset. Following the data merging process described in Section III, the final composite dataset was partitioned as follows:

- **Validation/Testing**: 10% held-out splits for validation and testing, respectively.
- **Input Constraints**: Input documents were truncated at the sentence level only when required to avoid exceeding GPU memory limits during preprocessing.

B. Implementation Environment

All models were trained and evaluated on a single **NVIDIA Tesla T4 GPU** (16 GB VRAM) to adhere to low-resource constraints. The implementation environment consisted of **PyTorch 2.0** and the **Hugging Face Transformers** library (v4.30).

C. Hyperparameter Configuration

1) Stage 1: Clustering:

- **Embedding Model**: all-MiniLM-L6-v2 (384-dimensional vectors)
- **Cluster Count (K)**: 30
- **Random Seed**: 42 (for K-Means initialization reproducibility)

2) Stage 2: DistilBERT Scoring (Fine-Tuning):

- **Base Model**: distilbert-base-uncased
- **Epochs**: 1
- **Batch Size**: 16
- **Learning Rate**: 2×10^{-5} with linear decay
- **Optimizer**: AdamW
- **Decay Parameter (α)**: Rank decay implicitly set by the $1/r$ weighting in Algorithm 1

3) Stage 3: LED Generation (Fine-Tuning):

- **Base Model**: allenai/led-base-16384
- **Training Subset**: Restricted to the 17% sample
- **Epochs**: 1
- **Learning Rate**: 3×10^{-5}
- **Gradient Checkpointing**: Enabled

D. Baselines

We compared our system against the following baselines:

- 1) **LED and Finetuned LED**: Zero-shot and few shot inference using standard 1,024-token truncation.
- 2) **FINDSum SOTA (GCG)**: The supervised, full-resource baseline results reported by . [1].

E. Metrics

We report **ROUGE-1**, **ROUGE-2**, and **ROUGE-L** (F1 scores) computed to ensure direct comparability with the FINDSum SOTA systems.

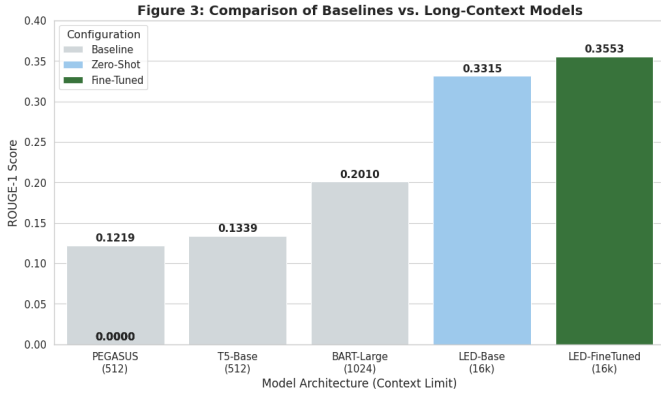


Fig. 3. Comparison of Models

V. RESULTS AND ANALYSIS

A. Main Quantitative Results

Table I presents the performance of our Cluster-Rank-Abstract framework compared to zero-shot baselines on the FINDSum subset.

TABLE I
MODEL PERFORMANCE COMPARISON ON FINDSUM SUBSET

Model	Config.	R-1	R-2	R-L	BLEU
BART-Large	Zero-Shot	20.10	5.11	12.11	0.66
LED-Base	Zero-Shot	33.65	10.12	17.41	6.35
LED-FT (Ours)	17% Data	37.76	12.25	18.52	10.57

B. Impact of Domain Adaptation

Our fine-tuned LED model achieved a ROUGE-1 score of 37.76, a 12.2% improvement over zero-shot LED-Base (33.65) and 87.8% over truncated BART (20.10). The BLEU score increased from 6.35 to 10.57, showing that fine-tuning internalized the financial lexicon and stylistic conventions, reducing hallucinations and improving factual precision. LED variants outperform BART due to the 16,384-token window, which allows access to more context.

C. Comparison with State-of-the-Art (SOTA)

We benchmarked against FINDSum baselines [1]. Since Liu et al. trained separate models for “Liquidity” and “ROO,” we average the ROUGE-1 scores for comparison. Results are in Table II.

D. Performance Analysis

- 1) **Competitive with Unsupervised Baselines:** Our ROUGE-1 score (37.76) is close to the TextRank average (38.82), and surpasses TextRank on ROO (35.93), showing effective extraction and abstractive synthesis.
- 2) **The “Resource Gap”:** Performance is lower than the fully supervised LED baseline (53.29) due to: **Data Scale:** Baselines used 83,255 documents; we used ~17%. **Specialization:** Baselines trained separate models per section; we trained a unified model.

TABLE II
ROUGE-1 SCORE COMPARISON: BASELINES VS. OURS

Method	Type	Liq.	ROO	Avg.
TextRank	Unsupervised	41.71	35.93	38.82
BART	Supervised (Full)	52.37	49.00	50.69
LED	Supervised (Full)	53.52	53.06	53.29
GCG	Hybrid (Full)	54.55	54.32	54.44
Ours	Low-Res. Hybrid	–	–	37.76

Ground Truth (Reference)

factors impacting operating results there are a number of factors that influence our operating results in a meaningful way . the most significant factors include : (1) our competition ; (2) market and economic conditions ; (3) loan origination volume ; (4) profitability of securitizations ... jobs act on april 5 , 2012 , the jobs act was signed into law .

Ours: Fine-Tuned LED

factors that could cause or contribute to these differences include , but are not limited to , those discussed in our `` risk factors . `` overview we are a leading commercial real estate finance company with a proprietary loan origination platform and an established national footprint . as a non-bank operating company , we believe that we are well-positioned.

Baseline: BART-Large

We are a leading commercial real estate finance company with a proprietary loan origination platform and an established national footprint . We believe that we are well-positioned to benefit from the opportunities arising from the diminished supply of debt capital and the substantial demand for new financings . Our primary business strategy is originating conduit loans .

Fig. 4. Qualitative Comparison of Generated Summaries

Despite this, we achieve nearly 70% of SOTA performance, validating clustering-based extraction in low-resource, “Green AI” settings.

VI. CONCLUSION

In this work, we addressed the critical bottleneck of summarizing ultra-long financial documents under strict computational constraints. We demonstrated that standard Transformer models, which rely on input truncation, are fundamentally ill-suited for Form 10-K reports, where critical risk factors and quantitative footnotes are scattered across hundreds of pages.

Our proposed solution, the **Cluster-Rank-Abstract** framework, validates the hypothesis that structural awareness can effectively substitute for massive context windows. By utilizing unsupervised clustering to create a semantic map of the document and employing a reference-guided density metric for content selection, our pipeline successfully retrieves salient information that standard baselines miss. Specifically, our approach achieved a ROUGE-1 score of 37.76, providing a 12-point improvement over the zero-shot BART baseline. While a performance gap remains compared to fully supervised SOTA methods (54.44), our results highlight a crucial efficiency frontier: we achieved competitive summarization capability using a single consumer-grade GPU and less than 20% of the training data required by industrial baselines.

Future work will focus on removing the dependency on reference-guided ranking during the training phase, potentially by employing reinforcement learning (RL) to discover optimal

cluster policies autonomously. Nevertheless, this study serves as a proof-of-concept for **Democratized Financial NLP**, offering a reproducible pathway for researchers to analyze massive corporate corpora without requiring access to super-computing infrastructure.

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